Classification of SSVEP Using a Subject-Independent Approach for Brain-Computer Interface Applications

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Abstract–This study investigates the feasibility of reliably classifying steady-state visually evoked potential (SSVEP) signals in a subject-independent setting using a support vector machine (SVM) classifier. Specifically, it evaluates the effectiveness of training the classifier exclusively on data from users other than the one whose data are used for testing. This approach aims to reduce or even eliminate the need for user-specific calibration, thereby increasing the practicality and scalability of brain–computer interface (BCI) applications in real-world settings. Using the proposed SVM method, an average classification accuracy of 78.49% was achieved across 15 subjects. Moreover, the SVM outperformed both canonical correlation analysis (CCA, 78.43%) and the harmonic-based method (60.49%). Notably, six subjects achieved accuracy above 90%, with the highest individual accuracy reaching 99.72%.

Keywords: SSVEP, SVM, subject-independent classification

I. Introduction

Steady-state visual evoked potentials (SSVEPs) are among the fundamental methods used in brain–computer interface (BCI) systems [1], [2]. They are elicited in electroencephalography (EEG) signals when a subject is exposed to a light source flashing at a constant frequency [3]. A major challenge in SSVEP-based BCIs is the variability of EEG signals, both across different subjects and between sessions of the same subject [4], [5], [6]. This variability complicates the design of reliable and generalizable BCI systems. Traditional approaches often rely on user-specific calibration, which is time-consuming and reduces the practical usability of BCIs. Consequently, there is growing interest in subject-independent classification methods that can perform effectively without requiring individualized adjustments.

Challenges in SSVEP-based brain–computer interfaces include inaccurate detection of stimulation frequencies and long calibration periods, both of which limit performance and contribute to increased eye fatigue during extended sessions [7]. An important research direction in BCI development is achieving interfaces that operate without session-specific calibration. Although calibration-free methods typically yield slightly lower accuracy, they offer the potential to simplify system use and enhance the practical utility of BCIs. Various approaches to calibration-free classification can be found in the literature. Recent efforts have focused on leveraging deep learning to automatically extract features from large datasets, thereby reducing the need for individual calibration [8]. Another widely adopted strategy is transfer learning, which uses data from a broader population to train models that can then be adapted to a new user with minimal or no additional data [9]. These methods aim to create highly generalizable models capable of performing effectively across users, environments, and sessions.

One solution optimized stimulus coding sequences and applied the advanced Filter Bank Canonical Correlation Analysis (FBCCA) algorithm to distinguish among a large number of targets while maintaining high accuracy and speed [10]. The system was evaluated in both offline and online experiments, achieving an average online accuracy of 87.16% and high information transfer rates. Notably, seven out of twelve participants achieved over 90% accuracy in real-time experiments. Another approach employed DG-Conformer networks based on transformer architectures [11]. This model integrates an enhanced transformer structure with domain generalization (DG) mechanisms to improve cross-subject generalization. Such improvements are crucial for SSVEP-BCI systems, as they reduce reliance on subject-specific calibration data. The proposed model also incorporates local convolutional modules to better capture the oscillatory characteristics of SSVEP signals. A further idea involves generating synthetic, subject-invariant signals [12]. The proposed method, termed Subject Invariant SSVEP Generative Adversarial Network (SIS-GAN), generates synthetic EEG signals that are unaffected by individual subject data. This is achieved by employing a pretrained classifier with fixed weights to ensure the generated data remain invariant to a given subject. Classifiers trained on these synthetic signals demonstrated substantially higher accuracy on previously unseen subjects, outperforming baselines by up to 16 percentage points.

II. Aim of the Article

The aim of this paper is to evaluate the performance of a calibration-free SSVEP signal classification system, in which the training process is carried out exclusively on data from users other than the one used for testing. The study employed a leave-one-subject-out (LOSO) strategy, where in each iteration the data of a single user were completely excluded from training and used solely for testing, while the data from the remaining participants were used to train the model. The results obtained with an SVM classifier using an RBF kernel are presented and compared with two reference methods: canonical correlation analysis (CCA), which matches measured EEG signals to sinusoidal and cosinusoidal signals at the stimulation frequencies and their harmonics, and a simple classification method based on selecting the stimulation frequency whose first harmonic exhibits the highest amplitude in the EEG spectrum. In addition, to assess the impact of calibration, classification performance in the calibration-free setting was compared with results obtained

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after adding a portion of the test user's data to the training set. This allowed us to determine the extent to which incorporating a limited amount of calibration data can improve classification accuracy.

III. Materials

The purpose of EEG recording in this study was to elicit steady-state visually evoked potentials (SSVEPs) through the use of visual stimuli at specific frequencies. A total of 15 recordings were conducted, one for each of the 15 participants. Each recording lasted 60 seconds and involved exposure to visual stimuli at frequencies of 7, 8, and 9 Hz. The stimuli were generated by an LED matrix measuring 65 × 65 mm, positioned 1 m from the participants' eyes. The brightness of the LEDs was adjusted to ensure comfort and to avoid visual fatigue. EEG signals were recorded using a g.USBAmp 2.0 amplifier (g.tec) with 24-bit A/D converters and a sampling rate of 256 Hz. Electrodes were placed on the scalp according to the international 10–20 system. Signals were acquired from electrodes O1, Oz, O2, Cz, and Fp1. To simplify data analysis and to provide an initial assessment of the proposed approach, only the signal from electrode Oz was used, as it is located in the occipital region and is particularly sensitive to visually evoked responses induced by SSVEP stimulation.

IV. Methods

EEG signals were preprocessed using a 4th-order Butterworth band-pass filter with a passband of 0.1–45 Hz. The purpose of filtering was to suppress low-frequency baseline drifts and high-frequency artifacts, including power-line interference. Furthermore, the data was normalized so that the mean was zero and the standard deviation was 1. No additional preprocessing methods, such as adaptive filtering, ICA, CAR, or biological artifact removal procedures, were applied. This approach enabled the evaluation of classifier performance under conditions of minimal interference with the input signal.

Feature extraction was then performed using spectral analysis with the Fourier transform. Each segment of 256 samples (1 second) was transformed via the FFT algorithm into 256 complex values. The magnitudes (amplitudes) were computed, yielding a frequency-domain representation of the signal with a resolution of 1 Hz.

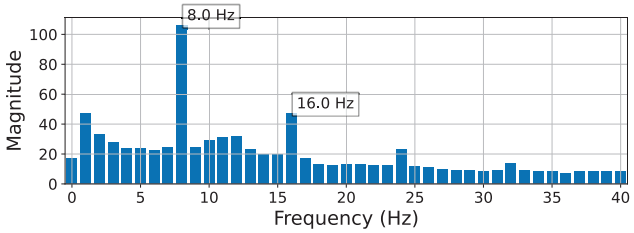


Fig. 1. Average frequency spectrum of Subject 12 for the 8 Hz stimulus.

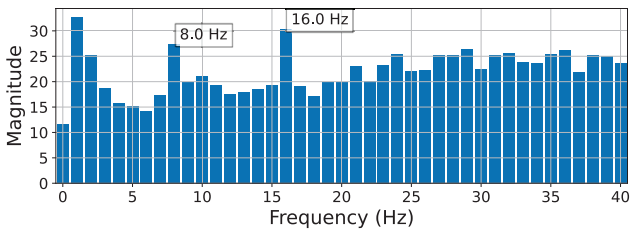


Fig. 2. Average frequency spectrum of Subject 9 for the 8 Hz stimulus.

Figures 1 and 2 illustrate the average frequency spectra of the analyzed signals for two representative cases. In both spectra, the x-axis corresponds to frequency in hertz (Hz), while the y-axis represents the magnitude of the frequency components. The spectra emphasize the fundamental low-frequency components at 8 Hz and 16 Hz, corresponding to the stimulus frequency and its second harmonic. However, clear differences emerge between subjects: the spectrum of Subject 12 shows a distinct peak at the stimulus frequency, indicating a more sharply defined frequency distribution, whereas the spectrum of Subject 9 exhibits a broader, more spread-out distribution. These examples highlight the variability across subjects, which is inherent to the characteristics of SSVEP signals.

For classification, a support vector machine (SVM) model was employed. SVM is a well-established machine learning method whose effectiveness has been demonstrated in numerous application domains [13]. Support vector machines are optimized using a quadratic loss function, allowing efficient error minimization. A key element of SVMs is the use of kernel functions, which map data into higher-dimensional spaces where classes can be linearly separated. Among the various kernels, the radial basis function (RBF) kernel is most widely used due to its strong performance in classification tasks. Although SVM is inherently a binary classifier, multiclass classification can be achieved using the one-versus-rest strategy. In this approach, multiple binary classifiers are trained, each distinguishing one class from all others, and their outputs are combined to produce the final prediction.

In this study, the SVM method was compared with two other standard classification techniques that do not require training examples of SSVEP:

1. Canonical correlation analysis (CCA) – This method computes canonical correlations between the measured EEG signal and a set of reference signals composed of sine and cosine waves at the stimulation frequencies and their second and third harmonics. For each stimulation frequency, the sum of correlations at the fundamental and harmonic frequencies is calculated. The frequency with the highest summed value is selected as the predicted class.
2. Harmonic amplitude method (Harm) – This method selects the stimulation frequency whose first harmonic exhibits the highest amplitude in the Fourier transform of the recorded EEG signal. The rationale behind this approach is that the dominant periodic component in the signal corresponds to the stimulus frequency, producing a distinct peak in the amplitude spectrum.

V. Results and Discussion

In the first approach to training the SVM classifier, a leave-one-subject-out (LOSO) strategy was applied. In each iteration, the training process was performed on data from 14 participants, while testing was carried out on data from the remaining participant. This ensured that the training set contained no samples from the test subject, allowing for a full assessment of the model's ability to generalize to unseen users. The procedure was repeated multiple times, each time designating a different participant as the test subject, so that each of the 15 participants served as the test set exactly once. This validation scheme provides a reliable evaluation of

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model performance in a calibration-free scenario, where the classifier is trained exclusively on data from other users and then applied to EEG signals from a subject whose data were not previously seen by the model.

Table I presents the classification accuracies obtained in the leave-one-subject-out procedure for three methods: the proposed SVM classifier with an RBF kernel, the harmonic amplitude method (Harm), and canonical correlation analysis (CCA). Each row of the table corresponds to one participant serving as the test set in a given iteration, while data from the remaining participants were used to train the models.

TABLE I. COMPARISON OF ACCURACY RESULTS OBTAINED FOR EACH SESSION LEFT FOR TEST IN CROSS-VALIDATION FOR CALIBRATION-FREE METHODS.

Subject	Classifier		
	SVM	*Harm*	*CCA*
S01	92,16	66,67	91,88
S02	86,27	63,59	87,39
S03	98,60	80,95	99,16
S04	90,48	76,47	90,76
S05	96,08	86,55	96,92
S06	71,71	36,41	57,14
S07	70,87	37,82	57,42
S08	68,63	59,38	71,99
S09	94,68	89,92	94,40
S10	57,98	44,54	57,42
S11	58,82	44,54	50,70
S12	61,06	38,94	64,15
S13	63,59	43,42	88,80
S14	66,67	43,98	68,35
S15	99,72	94,12	100,00
mean	78,49	60,49	78,43

The analysis of results shows that the SVM method achieved the highest average accuracy among all compared approaches, reaching 78.49%. In many cases (e.g., S01, S03, S05, S09, S15), classification accuracy exceeded 90%, and for participant S15 nearly perfect accuracy was achieved (99.72%). It is worth noting that even in cases with relatively lower accuracy (e.g., S10, S11), SVM still outperformed the Harm method by more than ten percentage points, demonstrating greater stability. The CCA method achieved an average accuracy similar to SVM (78.43%); however, for some participants (e.g., S06, S07, S10, S11), its performance was noticeably lower, suggesting greater sensitivity to individual differences in EEG signals. By contrast, the Harm method yielded the lowest average accuracy (60.49%) and the widest range of results, indicating that its usefulness in calibration-free scenarios is limited.

The next stage of analysis examined whether adding a small number of training examples from the test user could improve classification accuracy. To this end, the training set was expanded with the first 30% of examples from the current session of the test user, corresponding to 36 epochs obtained from the 60-second EEG recording. The resulting training set thus included data from all other participants as well as a portion of the recording from the test user, while the remaining 70% of examples from that session were used exclusively for testing. This approach made it possible to assess the extent to which limited calibration, based on a small number of examples from the target user, enhances the performance of the SVM classifier compared to the fully calibration-free variant. Table II presents the classification results obtained with the SVM classifier in two settings: (LOSO) a fully calibration-free scenario, and (LOSO+30%) a limited calibration scenario, in which the first 30% of examples from the test user's session were added to the training set. In both cases, the leave-one-subject-out procedure was applied, whereby in each iteration the data from one participant served as the test set, while data from the remaining participants (and, in the calibration scenario, a portion of the test user's session) were used for model training.

TABLE II. COMPARISON OF CLASSIFICATION ACCURACY FOR EACH SUBJECT USING SVM IN LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION WITH AND WITHOUT 30% CALIBRATION DATA.

Subject	SVM	
	LOSO	LOSO + 30%
S01	92,16	93,33
S02	86,27	84,76
S03	98,60	100,00
S04	90,48	93,33
S05	96,08	95,24
S06	71,71	78,10
S07	70,87	62,86
S08	68,63	79,05
S09	94,68	96,19
S10	57,98	59,05
S11	58,82	63,81
S12	61,06	64,76
S13	63,59	87,62
S14	66,67	67,62
S15	99,72	99,05
mean	78,49	81,65

The comparison of results shows that adding a small number of training examples from the test subject had a positive impact on classification performance in most cases. The average accuracy increased from 78.49% in the calibration-free scenario to 81.65% with limited calibration, representing an improvement of more than 3 percentage points. Improvements were observed in 11 out of 15 test sessions, while a decrease in accuracy occurred in 4 cases. Notably, in some sessions (e.g., S13 and S08), the improvement was particularly pronounced (24.03 and 10.42 percentage points, respectively), suggesting that adding a short segment of calibration data can substantially reduce classification difficulties for certain users. In no case did the calibration-free variant achieve substantially higher accuracy than the calibrated one, confirming that even limited calibration is either beneficial or at least neutral for classifier performance. Incorporating a fragment of the current session into the training set also improved classifier stability: both maximum accuracy (by about 3%) and minimum accuracy (by about 1%) increased, further highlighting the robustness of the model.

Additionally, feature importance was analyzed using the SHAP (SHapley Additive exPlanations) method, which allows for interpreting the contribution of individual features to the model's predictions [14]. This analysis made it possible to identify which features had the greatest impact on classification outcomes and how they contributed to the model's final decisions. Figure 3 shows the relative importance and influence of features for the model tested on subject S01. The vertical axis represents the frequency-domain features derived from the Fourier transform, while the horizontal axis shows SHAP values, indicating how each feature affects the model's prediction. The color of each point corresponds to the value of the given feature for a single instance, with the color scale displayed on the right-hand side

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of the plot. Red indicates high feature values, while blue indicates low values. The analysis reveals that the model's decisions were strongly influenced by the fundamental harmonics (7 Hz, 8 Hz, and 9 Hz) as well as the second harmonics (14 Hz, 16 Hz, and 18 Hz) of the stimulation frequencies, which is consistent with prior expectations. Importantly, the presence of harmonics among the most significant features enhances the interpretability of the model, as they directly correspond to well-known properties of the signal.

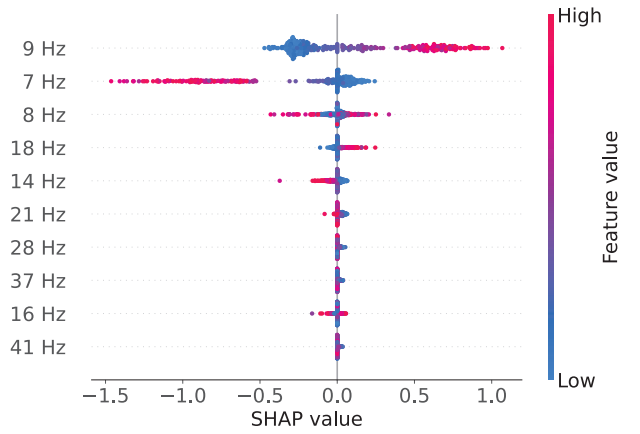


Fig. 3. Feature contribution plot based on SHAP values.

The SVM classifier achieved the best performance among all methods tested. Its advantage is not limited to high accuracy but also lies in its flexibility. In particular, the feature space can be easily extended by adding new features, and the SVM is able to incorporate them effectively during training. This property makes it suitable for problems where new information may gradually become available, potentially leading to further improvements in classification accuracy.

Another important strength of SVM is its optimization process: during training, the algorithm learns the optimal weights that define the separating hyperplane between classes. This contrasts with some other classifiers, such as k-Nearest Neighbors, which do not involve weight optimization but only rely on distance measures. As a result, SVM can better adapt to the structure of the data and generalize more effectively.

Furthermore, expanding the training dataset by including recordings from additional sessions significantly improves classification performance on unseen test data. SVM benefits from this because the optimization of its decision boundaries becomes more robust when trained on a richer and more diverse dataset. Taken together, these characteristics highlight why SVM is a strong choice for EEG-based classification tasks.

VI. Conclusions

The conducted experiments confirm that subject-independent classification of SSVEP signals can achieve high accuracy, making it possible in practice to eliminate the need for session-specific calibration. Based on the obtained results, it can be concluded that the SVM classifier with an RBF kernel is an effective tool for implementing calibration-free classification. These findings are highly relevant for the practical use of brain–computer interface (BCI) systems, as reducing or eliminating the calibration stage improves usability, shortens system setup time, and enables deployment in real-world conditions, including by non-technical users. Future research should focus on further improving the generalizability and robustness of such models across a broad range of users, diverse recording conditions, and session-to-session variability of EEG signals. An especially important direction is to extend experiments by incorporating a larger number of electrodes, which would allow more comprehensive capture of spatial information in EEG signals and potentially enhance classification accuracy. Additionally, combining classical methods with deep learning, optimizing models for real-time operation, and testing in mobile and immersive environments represent promising avenues for broadening the applicability of BCI technology.

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